

Utility of a Neurobehavioral Screening Battery for Differentiating the Effects of Two Pyrethroids, Permethrin and Cypermethrin

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McDANIEL, K. L. AND V. C. MOSER. *Utility of a neurobehavioral screening battery for differentiating the effects of two pyrethroids, permethrin and cypermethrin.* NEUROTOXICOL TERATOL 15(2) 71-83, 1993.—The ability of a neurobehavioral screening battery to differentiate the effects of two pyrethroids, permethrin and cypermethrin, was assessed in this experiment. Although the structures of these pesticides differ only in the α -cyano group, the behavioral syndromes associated with the Type I and II pyrethroids are quite different. The tests included a functional observational battery which is a series of subjective and quantitative measures of neurological function and behavior, and an automated measure of motor activity. Our results verified previous reports in the literature describing these different syndromes, i.e., aggressive sparring behavior, fine to whole-body tremor, hyperthermia, and decreased motor activity for the Type I pyrethroid permethrin, and pawing, burrowing, salivation, whole body tremor to choreoathetosis, hypothermia, and lowered motor activity for the Type II pyrethroid cypermethrin. In addition, we report that permethrin produced decreased grip strengths, increased resistance to capture, increased reactivity to a click stimulus, and induced head and forelimb shaking and agitated behaviors, whereas cypermethrin produced pronounced neuromuscular weakness and equilibrium changes, retropulsion, lateral head movements, alterations in responses to various stimuli, and increased urination. Although there were similarities in some effects (e.g., decreased motor activity), the pesticides differed sufficiently in their overall behavioral profiles, and severity and time course of effects, to discriminate these two compounds. Thus, this type of screening approach is sensitive enough to differentiate these pyrethroids for hazard identification purposes.

Pyrethroid	Permethrin	Cypermethrin
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THE use of rapid screening methods to assess the neurotoxic potential of chemicals has been recommended by expert panels (e.g., 16,31,46) and reviewed by the Office of Technology Assessment (41). This approach required by the U.S. Environmental Protection Agency for testing under the Federal Insecticide, Fungicide, and Rodenticide Act and the Toxic Substances Control Act (39,42), and is under consideration by the Organization for Economic Cooperation and Development (OECD) for inclusion in standard toxicity testing (OECD ad hoc Meeting on Neurotoxicity Testing; March 27-29, 1990). Furthermore, the validation and utility of neurotoxicity screening batteries has been the topic of workshops and symposia (e.g., 1,5,11,38,40). Ongoing research in our laboratory has been directed toward applying these methods to a broad variety of chemicals and testing conditions to assess how well such tests perform to our expectations. We are also developing a database of chemical effects against which the actions of unknown chemicals can be compared (21-28). Our neurobehavioral screening battery consists of a functional obser-

vatational battery (FOB), which is a series of tests to assess sensory, neuromuscular, and autonomic function, and an automated measure of motor activity. These tests are primarily intended to provide dose-response and time-course data, as well as profiles of neurological effects which can then be used to make decisions concerning a compound's neurotoxic potential and lead to subsequent studies for risk assessment (24).

In the present study we examined the ability of our neurobehavioral screening battery to characterize and differentiate the previously-described effects of two pyrethroids. Pyrethroids comprise a large class of widely-used pesticides, and the toxicity of pyrethroids in insects and vertebrates involves their effects on motor and sensory fibers in the central and peripheral nervous systems (for reviews see 4,13,45,47). The pyrethroids in general produce numerous neurobehavioral signs and alter the gating mechanism of the sodium ion channel in neural tissues (4,15,30,43,44). However, differences in the specific actions of these chemicals have been reported and

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these are summarized in Table 1. Pyrethroids have been classified as Type I or II depending on the particular behavioral syndrome they produce ('T' for tremors, or 'CS' for choreoathetosis and salivation, respectively; 15,43) and on their action on sodium ion channels (29,44). The major structural feature which differentiates the subgroups is that Type II pyrethroids usually contain an α -cyano moiety while the Type I pyrethroids do not (4,13).

In this experiment, we obtained dose-response and time-course data for the effects of permethrin (Type I pyrethroid) and cypermethrin (Type II) using our neurobehavioral screening battery, and profiles of effect on functional domains were established. The data obtained using this approach were shown to differentiate the actions of these two pyrethroids.

METHOD

Animals

Adult Long-Evans hooded rats of both sexes (Charles River Laboratories, Raleigh, NC) were used in all experiments and were 70–90 days of age when testing began. Rats used in the motor activity study were singly-housed in wire mesh cages, whereas for the FOB experiments rats were singly-housed in solid polycarbonate cages with hardwood chip bedding (Beta-Chip, Granville Milling Co., Creedmoor, NC). All rats were allowed free access to Purina rodent chow (Rat Chow No. 5002, Ralston Purina Co., St. Louis, MO) and deionized tap water in the home cages. The animal facility was maintained at $22 \pm 1^\circ\text{C}$, $55 \pm 5\%$ humidity, and had a 12 L : 12 D cycle with lights on at 0630 h. FOB testing took place in the animal housing room. For the motor activity tests rats were transported to another isolated room in the animal facility. Testing of each chemical was conducted within a time frame of 2 to 3 weeks, with the dose groups counterbalanced across days. When lethality occurred in the FOB studies, extra rats from within the same shipment were added to have the same number of rats per dose group. There were no available rats with which to replace those that died in the motor activity studies.

Chemicals

Technical grade permethrin (95% purity) and cypermethrin (97% purity) were kindly donated by Drs. D. Gammon and L. Smeltz (FMC Corp., Princeton, NJ). Both samples contained approximately equal quantities of *cis* and *trans* isomers. Chemicals were dissolved in corn oil (Mazola[®]) and administered by oral gavage in a volume of 1 ml/kg.

Functional Observational Battery

Procedure. The FOB consists of home-cage, handling, open-field, and manipulative behavioral measures, as well as physiological measures. The FOB protocol used in this study was based on a previous version (27). Since that initial report, however, experience with these tests has led to many modifications. For example, some FOB measures have been changed from descriptive to rank-order (such as the sensorimotor responses), and other measures (such as limb rotation, catalepsy, and extensor thrust) that were found to be insensitive to treatment or otherwise unreliable have been eliminated. Because fairly extensive changes have been made, the current protocol is described in Appendix 1.

The FOB tests were completed in approximately 6–8 min per rat. All data were recorded on standardized datasheets and later entered into a computer for further analysis. The observer was blind with respect to the dose level.

Experimental design. Eight rats of each sex were treated by gavage (po) with either corn oil (vehicle), 25, 75, or 150 mg/kg permethrin. For cypermethrin, test groups received either corn oil (vehicle), 20, 60, or 120 mg/kg. The selection of doses was based on motor activity dose-response data from Crofton and Reiter (8).

Preliminary studies using arousal and gait score assessments were conducted to determine times of peak effect for each compound on the day of dosing. It was observed that somewhat different effects were evident at different times after dosing; thus assessments were made at two time points on the day of dosing. These times were 2 and 4 h for permethrin, and 1.5 and 3 h for cypermethrin. In addition, rats were

TABLE 1
DIFFERENCES AND SIMILARITIES IN SELECTED BIOLOGICAL EFFECTS OF PYRETHROIDS*

	Pyrethroid	
	Type I	Type II
Behavioral syndrome	T syndrome: aggressive sparring, increased sensitivity to external stimuli, fine to whole body tremor, hyperthermia	CS syndrome: pawing, burrowing, salivation, coarse whole body tremor progressing to choreoathetosis
Motor activity	decreased	decreased
Acoustic startle response	increased amplitude, no effect on latency	decreased amplitude, increased latency
Operant performance	decreased responding	decreased responding, altered patterning
Action on Na ⁺ channel	repetitive firing; keeps channel in open state for msec	frequency-dependent depolarizing block; keeps channel in open state for minutes
Action on GABA system	none	inhibits GABA-induced chloride influx

*References: 2,7,8,10,12,13,15,17,32,33,43,44,45.

tested immediately prior to dosing (time 0) and at 24 and 48 h after dosing.

Data analysis. A severity scoring scheme was used to determine chemical effects on specific domains of neurological function (i.e., autonomic, activity, excitability, neuromuscular, and sensorimotor). This method of analysis normalized individual data for all measures to a 1-to-4 scale, where a score of 1 reflected what was often observed in control rats and a score of 4 reflected a rare occurrence in control rats (24,28). The conversion of individual data to a severity score was based on the data for control rats. Table 2 illustrates that for measures which produce continuous data, score assignments were based on the range defined by the mean and SD of the control group, whereas for the rank and quantal data the score assignments were based on the frequency of occurrence of each score in the control rats. Measures such as lacrimation, gait score, etc., were already defined in terms of severity of effect and no conversions were necessary.

For each rat at each time point, these severity scores were summed across the measures that comprised each functional domain. The composite scores were then averaged across treatment groups and analyzed across time using a repeated-measures analysis of variance (ANOVA). Because motor activity data were not collected in the same rats in these studies and motor activity is an important indicator of the activity domain, chemical effects on the activity domain were not analyzed.

Analysis of individual FOB measures within a functional domain was carried out as previously described (6,27). In addition, physiological measures such as body weight and body temperature were analyzed for each study. For all measures, two-way ANOVAs were carried out using a between-subject factor of dose and within-subject repeated measures across time. If the ANOVA produced a significant overall effect of dose or dose $\times$ time interaction, subsequent univariate analyses were conducted at each time point which included group comparisons to control values using Dunnett's *t* test. In all analyses, resulting *p* values ≤ 0.05 were considered significant.

Continuous data were analyzed by a linear model (GLM; 37) using each rat's time-0 value as a covariate. Descriptive and rank data were analyzed using a categorical procedure (CATMOD; 37). Due to the small dose-group size, analyses were conducted separately for males and females (6,27).

Motor Activity

Apparatus. Motor activity was measured in a maze composed of a series of interconnected alleys shaped like a figure-8 with two blind alleys projecting from the central arena (see ref. 36). Six phototransmitter/diode pairs were equally spaced around the figure-8 portion of the maze, and one pair was located in each of the blind alleys. Motor activity was recorded by a microprocessor as the number of photocell interruptions over a 1-h session.

Experimental Design. The motor activity experiments were designed to conform to the FOB studies, but a few modifications were necessary. Eight rats of each sex were dosed with either corn oil (vehicle), 25, 75, or 150 mg/kg permethrin. For cypermethrin the highest dose was lowered due to lethality noted in the FOB study and the resulting dose groups included corn oil (vehicle), 20, 60, or 100 mg/kg. Motor activity testing took place 4, 24, and 48 h after dosing for permethrin, and at 3, 24, and 48 h after dosing for cypermethrin. (Earlier time points of 1.5 and 2 h were excluded from the motor activity studies to avoid having two test sessions occurring so closely together.)

Data analysis. Total photocell counts over the session were subjected to a two-way ANOVA using a grouping factor of dose and repeated measures across time, followed by Dunnett's *t* test.

RESULTS

Lethality occurred during the testing of both compounds. During FOB testing, one male rat from the high dose groups of permethrin and cypermethrin died on the day of dosing (those rats were subsequently replaced using available extras).

TABLE 2
CRITERIA FOR CONVERSION OF FOB DATA TO SEVERITY SCORES

Severity Score	Measure		
	Continuous	Scalar	Other
1	within $X \pm 1 \times SD^*$	Score which $\geq 50\%$ receives	considered normal
2	within $X \pm 1.5 \times SD$	Score which > 1 but $< 50\%$ subjects receives	considered slightly deviant from normal
3	within $X \pm 2 \times SD$	Score which 1 rat receives, or is 1 rank from mode	considered somewhat deviant from normal
4	outside $X \pm 2 \times SD$	Score > 1 rank from mode	considered severely deviant from normal
	forelimb and hindlimb grip strength, landing foot splay, rears, activity counts, urination, defecation	handling reactivity, ease of removal, arousal, touch response, click response, tail pinch response, approach response	home-cage posture, lacrimation, pupil response, salivation, palpebral closure, clonic and tonic movements, gait score, righting reflex

*Within the range defined by mean $\pm$ SD. See refs. 24, 28.

Six out of 12 female rats that were dosed with 120 mg/kg cypermethrin died (of these, four were replaced, therefore $n = 6$ females at that dose for the FOB). During the motor activity tests, one male and three female rats administered 150 mg/kg permethrin died. A slightly lower dose of cypermethrin was used for the motor activity test (100 mg/kg instead of 120 mg/kg), and two male and no female rats died at that dose.

Overall, cypermethrin was the more potent compound since the low dose (20 mg/kg) had considerable activity on some measures, whereas the low dose of permethrin (25 mg/kg) had no detectable effects (no-observable effect level, or NOEL). The magnitude of effects seen with the high dose of permethrin (150 mg/kg) on some measures was similar to that for the middle cypermethrin dose (60 mg/kg).

Functional Changes

The pyrethroids significantly affected the excitability, neuromuscular, and sensorimotor functional domains in both male and female rats. The autonomic domain was also affected by cypermethrin in both sexes but was only marginally significant ($p < 0.042$) in male rats treated with permethrin. A summary of the severity scoring analysis and general changes in individual measures is presented in Table 3. Data for the FOB measures which were significantly altered by pyrethroid exposure are illustrated in Figures 1-5 or listed in Table 4. Most effects were obtained on the day of dosing, although a few residual effects (most notably neuromuscular) were still significant 1 to 2 days after dosing.

Close examination of the profiles of effect for these two compounds revealed both qualitative and quantitative differences. For example, permethrin produced aggressive behaviors, only affected a few of the neuromuscular measures, and

had no clear autonomic activity. On the other hand, cypermethrin affected gait, muscle function, tone, and equilibrium, and produced pronounced autonomic signs and burrowing behaviors. Female rats were generally more sensitive to the effects of cypermethrin, but there were no gender-related differences in the effects of permethrin.

Both compounds produced somewhat bizarre behaviors which have been described in the literature and the nature of these behaviors were very different. While in the home cage, rats dosed with permethrin were positioned with their backs against a corner, rearing up and assuming a boxing stance. Although these rats were individually housed, the postures met the description of "aggressive sparring" which has been reported for Type I compounds (43). They also displayed unprovoked agitated behaviors, i.e., dart around the cage and then retreat and cower in the corner or bounce on their haunches, and sometimes vocalize spontaneously. On the open field, permethrin-treated rats showed other unusual behaviors involving pronounced shaking of the head and forelimbs, slapping their own forelimbs, and the same unprovoked agitated reactions which were observed in the home cage.

In marked contrast, rats administered cypermethrin were often observed lying flat in the home cage with tremors. Shoveling or burrowing into the bedding was also observed, and this was more prevalent in females. In many cases, the rat's muzzle also appeared swollen. On the open field, these rats displayed excessive lateral head movements, a high incidence of retropulsion, and continued to shovel persistently with their noses (analogous to the burrowing behavior observed in the home cage).

The effects of cypermethrin on autonomic function in both sexes included profuse salivation. This initial sign of intoxication was significant only at 1.5 h after dosing (Table 4). Urina-

TABLE 3
PROFILE OF EFFECTS OF PERMETHRIN AND CYPERMETHRIN

Domain*	Cypermethrin	Permethrin
Autonomic	M:† 1.5, 3 h F:† 1.5, 3 h salivation increased urination	M: 4 h —
Excitability	M: 1.5, 3, 24 h F: 1.5, 3 h increased removal resistance decreased arousal choreoathetosis	M: 2, 4 h F: 2, 4 h increased removal resistance increased arousal whole body tremors increased handling reactivity
Neuromuscular	M: 1.5, 3, 24 h F: 1.5, 3, 24, 48 h splayed limbs, flattened posture decreased grip strengths altered righting increased landing foot splay	M: 2, 4, 24 h F: 4 h ataxia decreased grip strengths altered righting
Sensorimotor	M: 3h F: 3h increased click response decreased touch response decreased tail pinch response	M: 4, 48 h F: 4 h increased click response increased touch response increased approach response

*Could not include motor activity in functional domain, therefore analysis of severity scores not applicable.

†M: males; F: females.

For each functional domain, times during which that domain was significantly affected are indicated. Also listed are the general effects within each domain.

TABLE 4
EFFECTS OF CYPERMETHRIN AND PERMETHRIN ON MEASURES OF THE FOB WHICH WERE SIGNIFICANTLY ALTERED BY EITHER CHEMICAL

Dose (mg/kg)	Cypermethrin									
	Males					Females				
	0	20	60	120		0	20	60	120	
Salivation†										
1.5 h	1	1	1.5*	2.5*		1	1	2*	3*	
Urination†										
1.5 h	.13	.13	.63	.86*		.13	0	.38	.38	
Arousal†										
3 h	3	3	2.5*	2*		4	4	2.5*	2*	
Abnormal motor movements§										
1.5 h	0/8	1/8	6/8*	7/8*		0/8	0/8	4/8*	7/8*	
3 h	0/8	1/8	7/8*	7/8*		0/8	0/8	4/8*	8/8*	
24 h	0/8	0/8	2/8*	2/8*		0/8	0/8	0/8	1/6	
Forelimb grip strength (kg)†										
3 h ± SEM	1.012 ± .069	.973 ± .074	.654 ± .155*	.533 ± .152*		.993 ± .044	1.022 ± .053	.835 ± .114	.275 ± .051*	
Hindlimb grip strength (kg)†										
3 h ± SEM	.966 ± .054	.851 ± .034	.521 ± .066*	.511 ± .107*		.866 ± .033	.825 ± .054	.582 ± .063*	.379 ± .050*	
Landing foot splay (mm)†										
24 h ± SEM	66.1 ± 4.1	65.2 ± 4.0	81.6 ± 8.4*	80.8 ± 6.1*		59.3 ± 4.4	58.5 ± 3.9	62.4 ± 4.0	78.5 ± 4.5*	
Righting reflex†										
3 h	1	1	3.5*	4*		1	1	3*	4*	
Touch response†										
3 h	3	3	1.5*	1.5*		3	2*	2.5	1*	
Tail pinch response†										
3 h	4	4	1.5*	2*		3	2	4	1*	

continued

TABLE 4 Continued

Dose (mg/kg)	Permethrin									
	Males					Females				
	0	25	75	150	0	25	75	150		
Arousal†										
4 h										
Handling reactivity†	3	2*	2.5	3.5*	4	4	4	3		
2 h	2	2	2.5	2.5	2	2	2	2.5*		
Abnormal motor movements§										
2 h	0/8	0/8	0/8	1/8	0/8	0/8	0/8	4/8*		
4 h	0/8	0/8	3/8*	7/8*	0/8	0/8	3/8*	7/8*		
Forelimb grip strength (kg)†										
4 h	1.038 ± .048	1.021 ± .078	.739 ± .090*	.778 ± .039*	.774 ± .032	.771 ± .039	.789 ± .051	.599 ± .07*		
Hindlimb grip strength (kg)†										
4 h	1.178 ± .057	1.121 ± .085	.820 ± .080*	.751 ± .060*	.934 ± .063	.847 ± .058	.794 ± .060*	.737 ± .05*		
Righting reflex†										
4 h	1	1	1	2*	1	1	1	1		
Approach response†										
4 h	1.5	1.5	2	2	1.5	2	2	2*		
Touch response†										
2 h	3	3	3*	3.5*	3	2	2.5	2		

Data are provided for only the time at which the peak effect occurred. Dose groups statistically different from vehicle are indicated (*).

†Mean ± SEM; §Number of rats showing signs/number of rats in dose group.

tion was also increased in these rats on the day of dosing (males at 1.5 h; females at 3 h). Permethrin was without significant effect on any individual autonomic measure, even though the composite autonomic score showed a significant effect of treatment in male rats at 4 h. This was probably because the transformed severity scores for defecation were somewhat higher than controls; however, statistical analysis of the raw data was not significant.

Permethrin greatly increased excitability, whereas the effects of cypermethrin, though significant, were not as prominent. Ease of removal from the home cage (i.e., resistance to capture) was significantly affected by permethrin (both sexes) and cypermethrin (males only). However, the magnitude of this effect was much greater with permethrin. Scores for the ease of removal measure are shown in Fig. 1. Median scores are given since these scores are nonparametric. In this and all figures, dose-response and time-course data are shown for male rats; data for female rats in the high dose group are included for comparison. Most male rats administered 60 or 120 mg/kg of cypermethrin were rated as moderately difficult to remove (rank = 3, rat follows investigator's hand and does not sit quietly), whereas rats (male and female) dosed with 75 and 150 mg/kg permethrin appeared agitated, darted around

the cage, were very difficult to grab, and in a few cases, showed aggression toward the observer's hand (ranks of 5 or 6). Once out of the cage, however, cypermethrin-treated rats were no different to handle than vehicle-treated rats, whereas females in the high permethrin dose group showed an increased reactivity and resistance in being handled (handling reactivity, Table 4). Furthermore, permethrin-treated male rats in the high dose group appeared more alert, and darted about in the open field, in contrast to cypermethrin-treated rats which were less alert (arousal, Table 4).

Permethrin produced moderate to severe whole-body tremors, the incidence and severity of which were dose-related at 2 and 4 h after dosing (abnormal clonic movements, Table 4). Cypermethrin caused a combination of spasmodic, clonic movements and tonic extensions and flexions of the body musculature (choreoathetosis). This occurred at the two higher doses on the day of dosing and 24 h later (more so in males than females; Table 4).

Sensorimotor reactivity was also altered by both pyrethroids and these effects were transient, occurring at 3 h for cypermethrin and 4 h for permethrin (Table 4). For some stimuli, differences between the two compounds were evident in the direction of change for specific measures. Both pyrethroids significantly increased the response to the click stimulus, shown in Fig. 2. With permethrin, reactivity was uniformly increased, as evidenced by increased reactions to the touch stimulus (males only) and the approach of a pencil (females only). In contrast, responses to the touch and tail pinch stimuli were significantly decreased by cypermethrin (both sexes).

Both compounds produced gait abnormalities and weakness (i.e., decreased forelimb and hindlimb grip strength), but the effects of cypermethrin were more severe and persistent (Table 4). All of the neuromuscular changes were more prominent at the later time points on the day of dosing (3 h for permethrin, 4 h for cypermethrin). Gait scores are presented in Fig. 3. Cypermethrin significantly altered gait in all dose groups on the day of dosing, and the effects of the high dose were still significant at 48 h. The effects of cypermethrin on gait were described as ataxia and tip-toe walking that progressed to a combination of limb dragging and flattened posture. On the other hand, only the highest dose of permethrin produced an ataxic tip-toe gait. In male rats treated with either chemical, there was no differential sensitivity between forelimb and hindlimb grip strength. Furthermore, the middle and high doses of each compound produced approximately equal decreases (Table 4). In females, significant decreases in hindlimb grip strength were obtained at the middle and high doses, but only the high doses affected forelimb grip strength.

Cypermethrin but not permethrin also greatly affected equilibrium and muscle tone. Righting reflex was markedly impaired at 3 h in rats of both sexes (Table 4). Data for landing foot splay are incomplete for the day of testing, because most rats were too impaired to land properly and the test was not conducted on those rats. However, increased foot splay was significant in female rats in the high dose group at 24 and 48 h, and in male rats in the middle and high dose groups at 24 h. Permethrin produced slight alterations in righting reflex, but only in the high dose group (males), and landing foot splay was not affected at any time.

Motor Activity

Motor activity was markedly depressed on the day of dosing, and the effects of both pyrethroids on total counts are

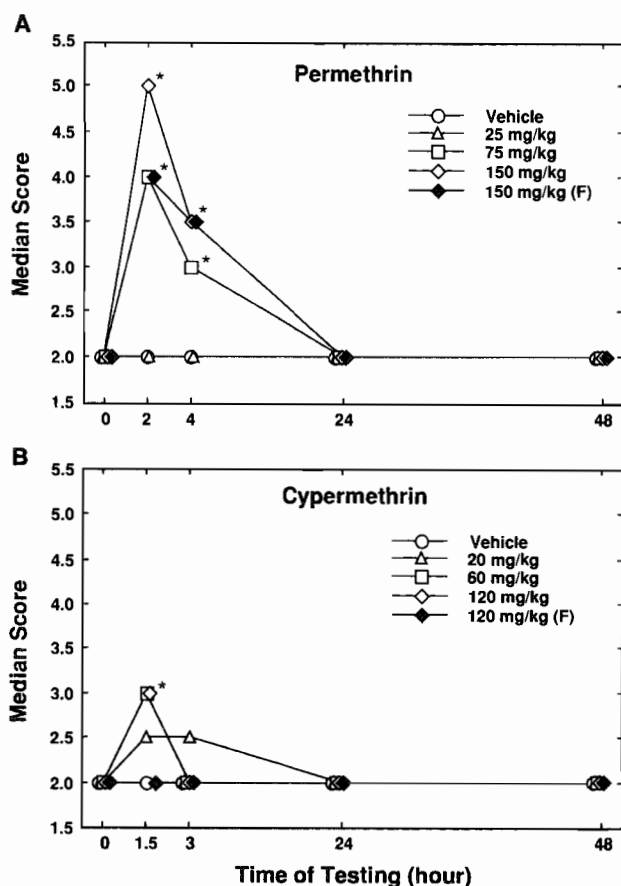


FIG. 1. Median score for ease of removal from the home cage for male rats treated with permethrin (A) and cypermethrin (B). Data for female rats treated with the high dose only are included. See Appendix for scoring criteria. $n = 8/\text{dose}$ for all dose groups, except $n = 6$ for females given cypermethrin 120 mg/kg. Dose groups statistically different from the appropriate control (i.e., male or female) are indicated (*).

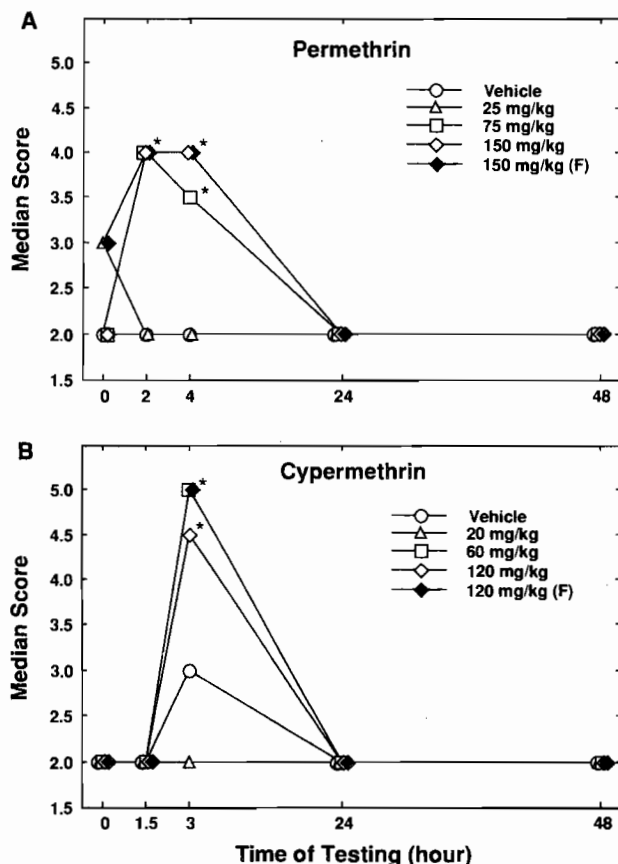


FIG. 2. Median score for the rating of the click response for male rats treated with permethrin (A) and cypermethrin (B). Data for female rats treated with the high dose only are included. See Appendix for scoring criteria. $n = 8$ /dose for all dose groups, except $n = 6$ for females given cypermethrin 120 mg/kg. Dose groups statistically different from the appropriate control (i.e., male or female) are indicated (*).

presented in Fig. 4. At 4 h after permethrin administration, the middle and high doses significantly decreased activity counts. Total counts in the high dose group were 39% and 16% of vehicle control values for males and females, respectively. Recovery was evident at 24 and 48 h, and activity was even significantly increased in the female high dose group at 48 h. At 3 h after dosing with cypermethrin, all doses significantly decreased activity in both sexes, and the high dose was still effective at 24 h. Furthermore, the low dose (20 mg/kg) was close to the ED_{50} value for this measure, in that it decreased activity by 46% and 43% in males and females, respectively. No effects of cypermethrin were obtained at 48 h.

Physiological Measures

Body weight loss was produced by cypermethrin in both sexes, and the effect of the high dose was still evident at 48 h. Permethrin produced only a transient weight loss in male rats on the day of dosing. The magnitude of the weight loss was greater with cypermethrin, with rats in the high dose group losing a maximum (at 24 h) of 7% and 9% of the predosing weight in male and female rats, respectively. Male rats dosed with the high dose of permethrin lost a maximum of 4% body

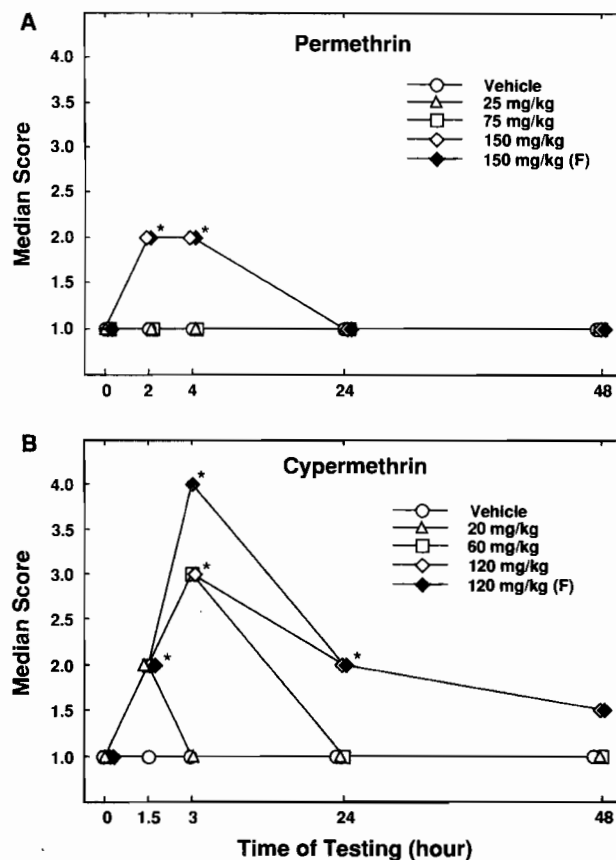


FIG. 3. Median score for the rating of gait abnormality for male rats treated with permethrin (A) and cypermethrin (B). Data for female rats treated with the high dose only are included. See Appendix for scoring criteria. $n = 8$ /dose for all dose groups, except $n = 6$ for females given cypermethrin 120 mg/kg. Dose groups statistically different from the appropriate control (i.e., male or female) are indicated (*).

weight (at 4 h), and the 3% weight loss recorded in females was not statistically significant.

Body temperature was markedly increased by the two higher doses of permethrin on the day of dosing, as shown in Fig. 5. Cypermethrin produced hypothermia in both sexes. In addition, the low dose of cypermethrin (20 mg/kg) produced a small but significant increase ($<1^{\circ}\text{C}$ higher than vehicle) in temperature at 1.5 h (male rats only).

DISCUSSION

Differences and similarities between permethrin and cypermethrin were evident by their effects on a neurobehavioral screening battery. Behavior representing all of the functional domains assessed were affected by both compounds, indicating their broad neurological activity. These effects were generally seen in both sexes with peak effects occurring within hours of dosing. Differences between the two chemicals emerged when their effects on individual measures of each domain were compared in terms of (a) qualitative signs of toxicity, (b) magnitude of effect, and (c) time course and effective dose range.

As in previous reports (e.g., 13,15,43), detailed descriptions of pyrethroid-induced clonic motor movements and bi-

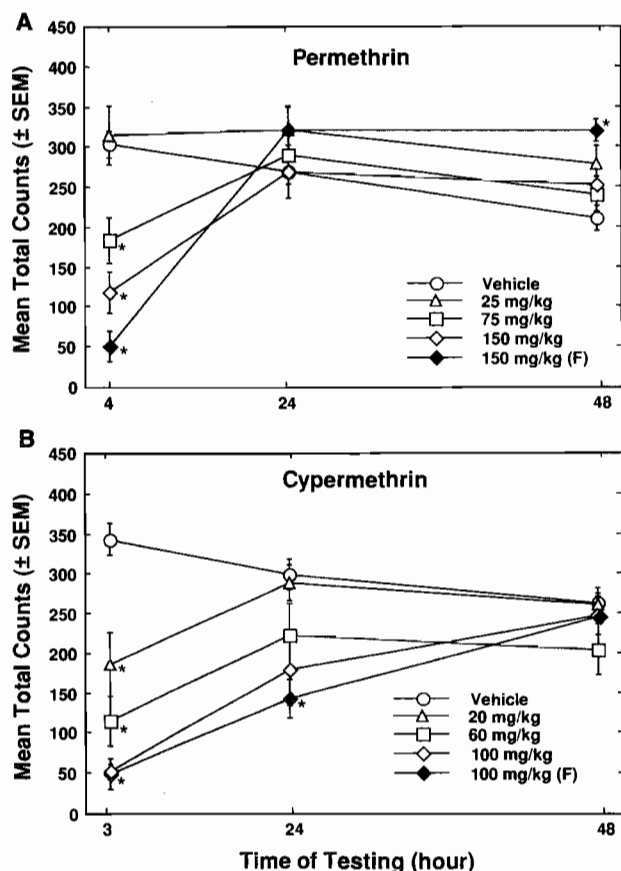


FIG. 4. Mean ($\pm$ SEM) total activity counts during a 1-hour session for male rats treated with permethrin (A) and cypermethrin (B). Data for female rats treated with the high dose only are included. $n = 8$ /dose for all dose groups, except: $n = 6$ for males given cypermethrin 100 mg/kg, $n = 7$ for males and $n = 5$ for females given permethrin 150 mg/kg. Dose groups statistically different from the appropriate control (i.e., male or female) are indicated (*).

zarre behaviors served to differentiate the syndromes produced by these chemicals. These included whole body tremor and aggressive sparring behavior after permethrin (a Type I pyrethroid) administration, and pawing, burrowing, and tremor progressing to choreoathetosis after cypermethrin (a Type II pyrethroid). Permethrin-treated rats displayed postures resembling sparring behavior even though they were individually housed. Head shaking and slapping of the forelimbs were also observed effects of permethrin, and these behaviors are similar to the shaking of head and forelimbs reported with intraventricularly-administered Type I pyrethroids (14). After cypermethrin administration, rats displayed pawing and burrowing behavior even while on the open field, indicating that these may be stereotypic rather than "purposeful" behaviors. The swollen muzzles, which were quite prominent after cypermethrin exposure, may have been a direct effect of the chemical or due to irritation possibly produced by the excessive burrowing actions. It is interesting to note that, in our experience, these pyrethroids are the only compounds which induce spontaneous vocalizations.

Only anecdotal accounts have been reported in the literature for some of the behavioral changes produced by pyrethroids. In this study, rating scales for assessing these

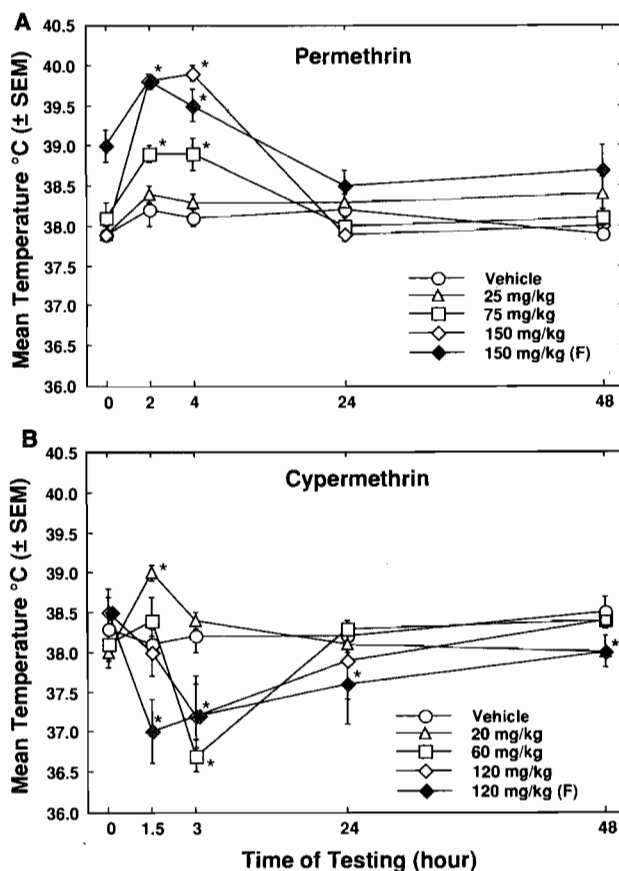


FIG. 5. Mean ($\pm$ SEM) body temperature data for male rats treated with permethrin (A) and cypermethrin (B). Data for female rats treated with the high dose only are included. $n = 8$ /dose for all dose groups, except $n = 6$ for females given cypermethrin 120 mg/kg. Dose groups statistically different from the appropriate control (i.e., male or female) are indicated (*).

behaviors provided semiquantitative data which exhibited dose-response and allowed statistical analyses to be conducted. These behaviors included those indicative of: increased sensitivity to external stimuli (seen for both types of pyrethroids); ataxia (permethrin); splayed hind limbs and abnormal locomotion (cypermethrin). For example, ataxia and abnormal locomotion were ranked using the gait and mobility scores; splayed hind limbs were quantified using the landing foot spread test. Reactivity to various stimuli (specific and nonspecific) was quantified by scoring the reactivity to being removed from the cage, and the responses to the tail-pinch, click, touch, and approach stimuli. Analysis of these measures also revealed differences between the effects of these compounds. For instance, permethrin markedly increased resistance to removal, whereas cypermethrin produced only a small effect on this measure (Fig. 1). These data probably indicate that permethrin generally increases overall excitability, since other sensorimotor responses were also increased. In contrast, cypermethrin generally decreased sensorimotor reactivity, with the exception of the click response which was increased.

Both chemicals produced neuromuscular changes as evidenced by decreased grip strengths, gait changes, and altered righting ability. However, the effects of cypermethrin were

more severe and persistent (e.g., see Fig. 3). Cypermethrin, but not permethrin, decreased muscle tone as evidenced by splayed legs, increased landing foot spread, and flattened posture. These differences were not simply a matter of potency and/or the doses utilized. In comparing the effects of the high dose of permethrin (150 mg/kg) with those for the middle dose of cypermethrin (60 mg/kg), which were approximately equi-effective on hindlimb grip strength (38% and 43% decrease for permethrin and cypermethrin, respectively), cypermethrin produced a greater decrease in forelimb grip strength (Table 4), greater degree of gait abnormality (Fig. 3), and in addition significantly altered righting, impaired mobility, and increased foot splay (Table 4). Although deltamethrin has been shown to increase landing foot splay (14), we could not find literature reports of the pronounced neuromuscular changes which we present here for cypermethrin.

The behavioral effects of both pyrethroids were significant at both the middle and high doses. Neuromuscular dysfunction produced by cypermethrin was evident at even the low dose. Most previous, purely-descriptive studies of the behavioral effects of these pesticides have employed only near-lethal doses (3,34,43). This provides evidence that behavioral evaluations conducted using standardized rating and testing criteria are sufficiently sensitive to evaluate behavioral effects of chemicals at lower doses, such as those used in first-tier screening studies. The behavioral data reported here generally show cypermethrin to be more potent than permethrin, and indeed addition of the α -cyano group has been shown to increase the potency of pyrethroids (13).

Reported LD50 values in rodents for cypermethrin are around 250–300 mg/kg (19,35). However, in the FOB study the high dose (120 mg/kg) produced 11% lethality in male rats and 50% in females. Although there is no previous evidence of this suggested gender difference in sensitivity, most studies have not compared males and females and reported LD50 values often do not specify sex. Lethality data for permethrin are difficult to derive from the literature, and LD50 values in rodents range from 410 mg/kg (35) to >5 g/kg (18). These differences may be due to isomeric potency differences. For permethrin, the *trans* isomer is not neurally active (4,13), and

batches of permethrin may vary in the ratios of *cis* and *trans* isomers. Thus, comparison of effective dose ranges of permethrin between studies can be misleading.

Though not statistically tested, there were differences in time course of effects between permethrin and cypermethrin. For cypermethrin, two phases of toxicity were evident. Salivation and increased removal reactivity (males) were most evident at 1.5 h. By 3 h these signs were subsiding, and the pronounced motor and sensory effects were apparent (e.g., compare the time course in Fig. 1 with those in Figs. 2 and 3). It is possible that different mechanisms are acting to produce these differences in time course. For permethrin, most effects were maximal at 4 h, or else they were approximately equal at 2 and 4 h. There were also differences in the duration of action of the two pyrethroids. Rats dosed with permethrin recovered by 24 h, but in those given cypermethrin several motor effects (e.g., gait changes, lowered grip strengths, decreased motor activity) were still evident 1–2 days later (especially in females).

The profiles of the neurobehavioral effects produced by these two structurally similar compounds were quite different. The signs of toxicity reported in this study correspond well to those of earlier reports. In addition, we report here pronounced neuromuscular changes produced by cypermethrin, and increased excitability and reactivity produced by permethrin. Thus, the neurobehavioral test battery clearly detected neurological activity, defined dose-response and time-course characteristics, and provided additional information for differentiating the distinct syndromes produced by these two pyrethroids.

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Appendix 1

Protocol for conducting a functional observational battery (modified from Moser et al., ref. 25).

HOME-CAGE MEASUREMENTS CARRIED OUT WHILE RAT IS IN HOME CAGE

Posture (Descriptive)

1. Sitting or standing
2. Rearing
3. Asleep, lying on side or curled up
4. Flattened, limbs may be spread out
5. Lying on side, limbs in air
6. Crouched over
7. Head bobbing

Note: Only 1, 2, or 3 are "typical"

Involuntary Motor Movements (Descriptive)

Clonic

1. Repetitive movements of mouth and jaws

2. Quivers of limbs, ears, head, or skin (sometimes seen in untreated rats)
3. Mild tremors
4. Severe or whole body tremors
5. Myoclonic jerks
6. Clonic convulsions
7. Wet dog shakes

Tonic

1. Contraction of extensors such that limbs are rigid and extended
2. Opisthotonus: head and body rigidly arched backward
3. Emprosthotonus: head and body rigidly extended forward
4. Explosive jumps into the air with all feet leaving the surface
5. Severe clonic and/or tonic convulsions resulting in dyspnea, postictal depression, or death

Vocalizations (Quantal):

Spontaneous, not in reaction to being handled. Also includes spontaneous vocalizations in the open field.

Palpebral Closure (Ranked)

1. Eyelids wide open
2. Eyelids slightly drooping
3. Eyelids drooping approximately half-way
4. Eyelids completely shut

REMOVE RAT FROM CAGE

Ease of Removing Rat From Cage (Ranked)

1. Very easy (rat sits quietly, allows investigator to pick it up)
2. Easy (vocalizations, without much resistance to being picked up)
3. Moderately difficult (rat rears, often following investigator's hand)
4. Rat flinches (with or without vocalizations)
5. Difficult (runs around cage, or is hard to grab, with or without vocalizations)
6. Very difficult (tail and throat rattles, with or without vocalizations)

Reactivity to Being Handled (Ranked)

1. Low (no resistance, rat is easy to handle)
2. Moderately low (slight resistance to being handled, with or without vocalizations)
3. Moderately high (rat may freeze, or be tense, or rigid in hand, with or without vocalizations)
4. High (squirming, or twisting, or attempting to bite, with or without vocalizations)

MEASUREMENTS MADE WHILE HANDLING RAT

Remove from cage and hold in hand. Note (under "Other" on data sheet) such things as increased or decreased body tone, bite marks, soiled fur appearance, missing toe nails, piloerection, emaciation (shallow stomach, prominent spinal vertebrae), or death. (Observations such as piloerection may also be made while rat is on open field)

Lacrimation (Ranked)

1. None
2. Slight
3. Severe

Palpebral Closure

Same as in home-cage measurements.

Salivation (Ranked)

1. None
2. Slight
3. Severe

Piloerection (Quantal)

"+" indicates presence of piloerection (i.e., coat does not lie down after stroking).

OPEN-FIELD MEASUREMENTS

Rat is placed in the center of a flat surface with a perimeter barrier covered with clean absorbent paper for exactly 3 min. During this time, the number of rears is counted and other

observations are made. (Suggested size of cart: approximately 60 × 90 cm with a 6.5 cm rim):

Rearing (Count)

Defined as each time the front legs of the rat come completely off the surface, although the rat does not necessarily have to raise itself up (i.e., this is a measure of the ability of the rat to place its weight on its haunches). Includes when the rat uses the side or lip of a cart top as support.

Involuntary Motor Movements

Same as in home-cage measurements.

Gait (Descriptive)

Note, if rat did not move during the 3-min observation period, it may be gently prodded (after the 3 min is over) in order to observe the gait.

1. Ataxia, excessive sway, rocks, or lurches
2. Hindlimbs show exaggerated or overcompensated movements, drag, or are splayed
3. Feet markedly point outward from body
4. Forelimbs drag, are extended, or unable to support weight
5. Walks on tiptoes
6. Hunched or crouched body position
7. Body drags or is flattened against surface

Gait Score (Ranked)

Ranking of gait abnormalities.

1. No abnormal gait
2. Slightly abnormal
3. Moderately abnormal
4. Severely abnormal

Mobility Score (Ranked)

Ability of rat to locomote despite gait abnormalities (different from gait score)

1. No impairment
2. Slightly impaired
3. Somewhat impaired
4. Totally impaired

Arousal (Ranked)

Level of unprovoked activity and alertness in the open field

1. Very low (stupor, coma)
2. Low (somewhat sluggish, some head or body movement)
3. Somewhat low (slightly sluggish, some exploratory movements with periods of immobility)
4. Alert, exploratory movements
5. Somewhat high (slight excitement, tense, excited, sudden darting or freezing)
6. Very high (hyperalert, excited, sudden bouts of running or body movements)

Stereotypy

Record any behaviors that are excessive or repetitive such as circling, stereotypic grooming, pacing, repetitive sniffing, or head weaving.

Bizarre Behavior

Record any unusual behaviors such as self-mutilation, Straub tail, retropulsion, writhing, flopping.

Excretion

At the end of 3 min, measure defecation and urination.

Defecation (Count)

Number of fecal boluses on paper. "D" will be recorded if diarrhea is present

Urination (Count)

Number of pools of urine on the paper. "X" will be recorded if polyuria, or overlapping pools, is present

STIMULUS REACTIVITY

Performed while rat is sitting on cart surface.

Approach Response (Ranked)

Approach rat head-on with the end of a blunt object, such as a pencil, hold approximately 3 cm from face for 4 s

1. No reaction
2. Rat slowly approaches and sniffs or turns away
3. Rat flinches, actual muscle contractions
4. More energetic response than 2) or 3)
5. Exaggerated reaction—jumps, bites, or attacks

Touch Response (Ranked)

Coming in from the side, touch rump gently with blunt object, such as a pencil

1. No reaction
2. Rat may slowly turn or walk away, or vocalizations with little or no movement
3. Rat flinches, actual muscle contractions
4. More energetic response than 2. or 3)
5. Exaggerated reaction: jumps, bites, or attacks

Click Response (Ranked)

Position clicker approximately 5 cm above the back of the rat and make sudden sound.

1. No reaction
2. Slight reaction, some evidence that noise was heard
3. Rat flinches, actual muscle contractions
4. More energetic response than 2) or 3)
5. Exaggerated reaction—jumps, bites, or attacks

Tail Pinch Response (Ranked)

Metal tweezers are used to squeeze the tail approximately 2–3 cm from the tip

1. No reaction
2. Rat may turn or walk forward, or vocalizations with little or no movement
3. Rat flinches, actual muscle contractions
4. More energetic response than 2 or 3
5. Exaggerated reaction: jumps, bites, or attacks

Pupil Response (Quantal)

The beam of a penlight flashlight is brought in from the side of the rat's head. Constriction of the pupil is noted with a "+", and "–" indicates lack of response. (This may be difficult to observe in some strains of rats and may be dependent on ambient lighting conditions)

Righting Reflex (Ranked)

Rat is held supine, then dropped from approximately 30 cm. Score ease of landing. Note, if the rat is paralyzed or severely affected, this test and the landing foot splay will not be carried out so as not to injure the rat.

1. Rat lands on feet
2. Slightly uncoordinated
3. Lands on side
4. Lands on back

Forelimb and Hindlimb Grip Strength (Continuous)

Strain gauges are used with wire mesh screens for the rats to grab. Screen for the forelimb grip measurement is oriented horizontally from the strain gauge, while the hindlimb grip screen is placed at a 45 angle from the gauge to allow full contact with the hind feet when the rat is pulled off the support platform. Two readings are taken and averaged (modified from Meyer et al., ref. 9).

*Body Weight (Continuous)**Body Temperature (Continuous)*

Rectal temperature is taken and thermister allowed to stabilize before reading

Landing Foot Splay (Continuous)

Fourth digit pads of hind feet are dotted with Tempra paint. Rat is dropped twice from prone position 40 cm from paper, and ink spots where he lands are noted. Measure distance between middle of ink blots, and average (modified from Edwards and Parker, ref. 20).

Other

Includes torn toenails, broken teeth, soiled fur, fur discoloration, convulsions at any time other than in the home cage or open field, crustiness around face or eyes, red pigmented excretions from eyes, or any findings which may impact the data.

